Building Science

Lecture 5: Thermal Comfort

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Lecture outline

- Understanding Thermal Comfort
- Achieving Thermal Comfort
- Psychrometric introduction

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Building Science

Course content

1. General introduction to Building Science
2. Climate impacts on Building orientation
3. Energy efficiency and Passive control strategies
4. Thermal performance – heat transfer principles
5. Thermal comfort
6. Thermal performance – Insulation and thermal bridges

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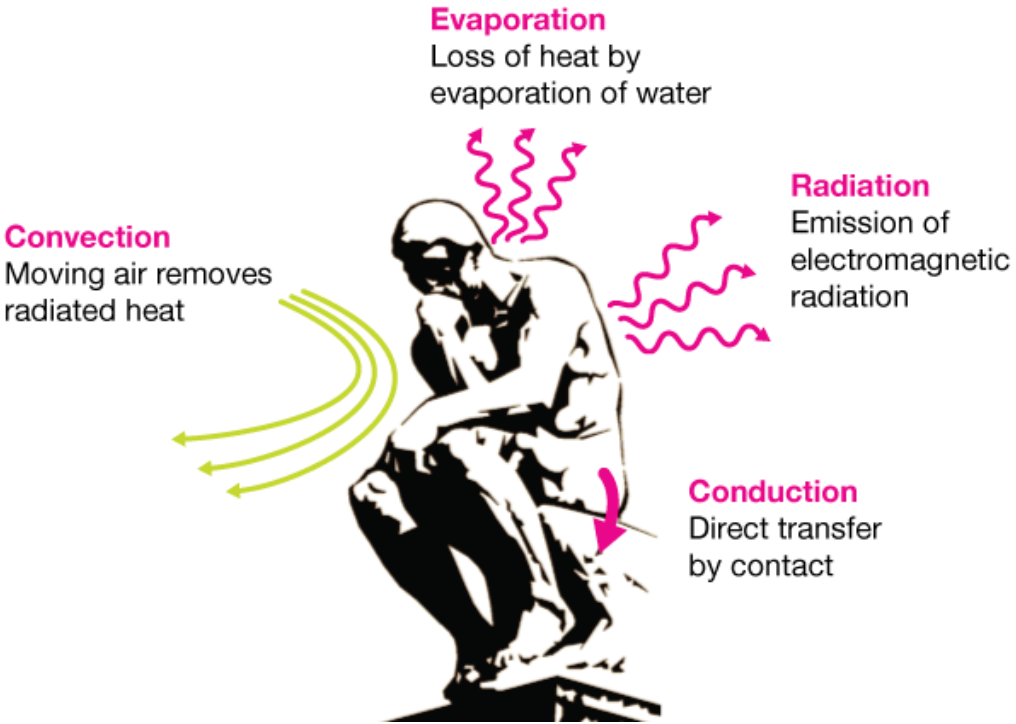
Thermodynamics

Heat Flow



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Thermal Comfort

What are the variables to achieve comfort for people in buildings?

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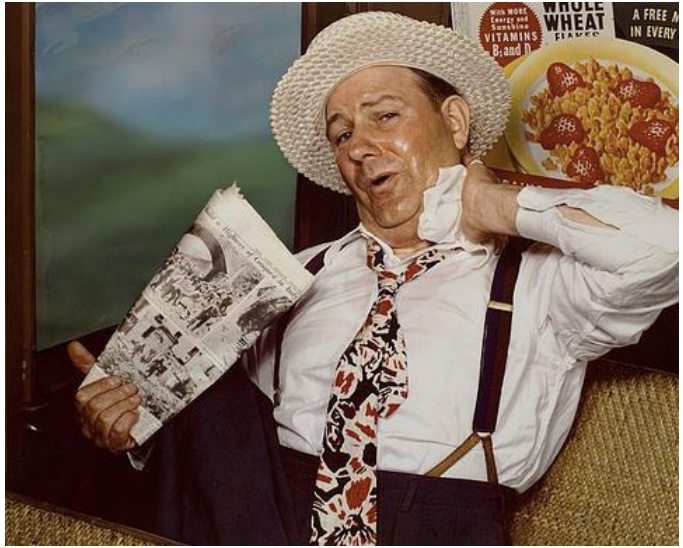
Thermal Comfort

What is it?

✓We all know what it's like to be *too hot* or *too cold*

✓Buildings are created to *protect us* from *external environments*



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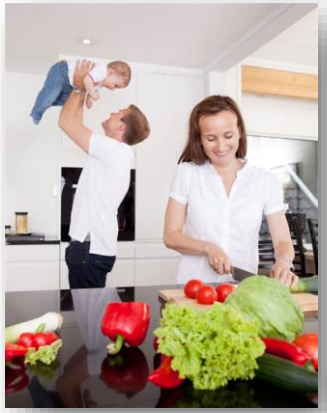
Thermal Comfort

Definition:

- ✓The *absence* of irritation or discomfort due to heat or cold, and
- ✓A human state involving *pleasantness of sensation*



— "*the condition of mind that expresses satisfaction with the thermal environment, and requires subjective evaluation*"

(ANSI/ASHRAE Standard 50)

This definition embraces factors beyond the physical and physiological

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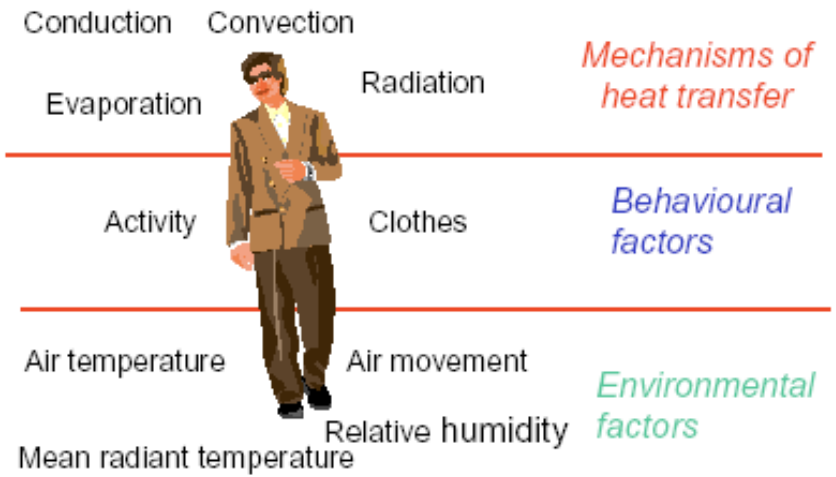
Thermal Comfort

Factors in thermal comfort



Environmental	Personal	Contributing factors
Air temperature	Metabolic rate (activity)	Food and drink
Radiant temperature	Clothing	Body shape
Relative humidity	State of health	Subcutaneous fat
Air movement	Acclimatisation	Age and gender

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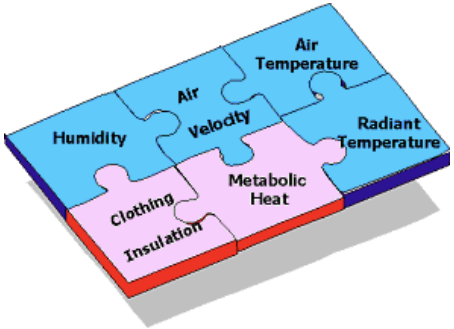
Thermal Comfort



Six factors

- ✓Air temperature (∘C) — Temperature of the air surrounding the occupant
- ✓Radiant temperature (∘C) — The weighted average of all the temperatures from surfaces surrounding an occupant
- ✓Relative humidity (%RH) — Percentage of water vapour in the air
- ✓Air movement. (m/s) — Rate of air movement given distance over time
- ✓Metabolic rate (MET) — The energy generated from the human body
- ✓Clothing insulation (CLO) — The amount of thermal insulation the person is wearing

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Thermal Comfort

Predicted Mean Vote (PMV)

- ✓ A thermal scale that runs from Cold (-3) to Hot (+3) – originally developed by Professor P.O. Fanger and later adopted as an ISO standard.
- ✓ The original data was collected from a large number of people experiencing different conditions within a climate chamber.
- ✓ A mathematical model of the relationship between all the environmental and physiological factors was developed
- ✓ The recommended acceptable PMV range for thermal comfort from ASHRAE 55 is between -0.5 and +0.5 for an interior space

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Thermal Comfort

Predicted Mean Vote (PMV)

PMV sensation scale

Value	Sensation
-3	Cold
-2	Cool
-1	Slightly cool
0	Neutral
1	Slightly warm
2	Warm
3	Hot

Source: http://academy.autodesk.com/library/building-science/human-thermal-comfort

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Thermal Comfort

Predicted Percentage of Dissatisfied (PPD) occupants

- ✓ Predicts the percentage of occupants that will be dissatisfied with the thermal indoor conditions
- ✓ A function of PMV, given that as PMV moves further from 0, or neutral, PPD increases
- ✓ The recommendation of acceptable PPD range for thermal comfort from ASHRAE 55 is less than 10% persons dissatisfied for an interior space

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Achieving Thermal Comfort

How can buildings provide people comfort?

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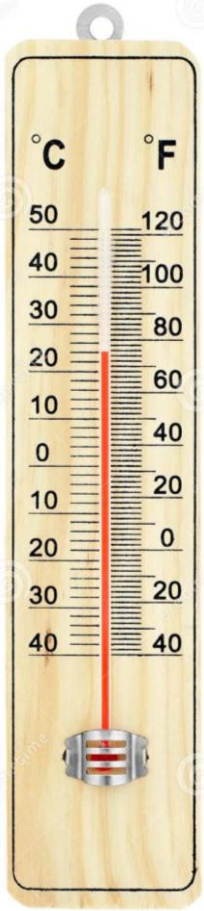
Achieving Thermal Comfort

Air temperature
(also known as ‘Dry Bulb Temperature’)

✓Air temperature, °C:

- Is the dominant environmental factor
- Is impacted by acclimatisation
- Subjective non-climate factors may impact perceptions (e.g. emotions)
- Around 19-24 ∘C is good (depending on acclimatisation)



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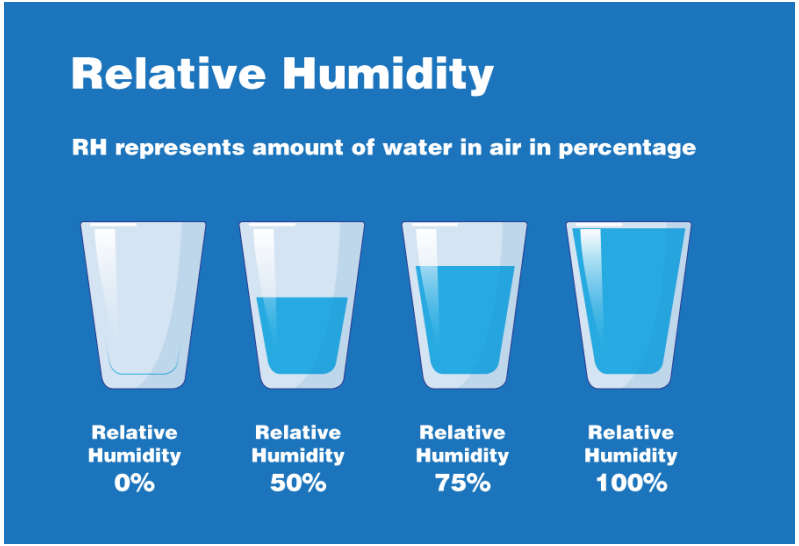
Achieving Thermal Comfort

Relative humidity (%RH)

✓Relative humidity:

- is not as important at comfortable air temperatures
- Usually not a problem for human comfort if kept between 20% and 80% RH.
- At DBT>28∘C, humidity important because sweating starts



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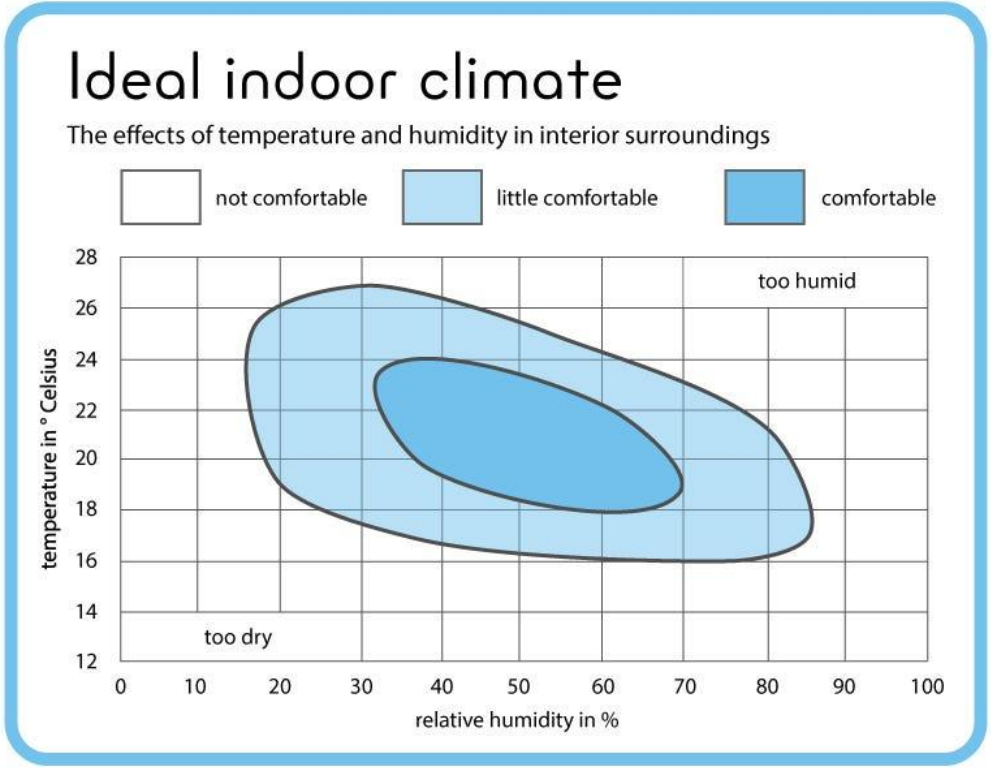
Achieving Thermal Comfort

Relative humidity (%RH



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Relative humidity (%RH) (cont'd)

✓In Summer:

- high %RH stops good evaporation from the skin and in respiration (breathing) – therefore it *feels hotter*.

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Relative humidity (%RH) (cont'd)

✓In Winter:

- Very low RH leads to dry skin and throat
- Skin problems (e.g. Eczema)
- Can also lead to easier transmission of respiratory diseases

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Air movement

✓Air movement (or air velocity):

- Affects convective loss or gain
- Increases evaporation (RH, 40-50%)
- Is not as important if RH is low (<30%)
- Can cause *draft* in cold conditions
- If insufficient, can feel ‘stuffy’, or ‘dank’

<0.1 m/s	Stuffy
To 0.2	Unnoticed
To 0.5	Pleasant
To 1	Awareness
To 1.5	Draughty
>1.5	Annoying

Subjective reactions to air movement Source: Szokolay, 2008

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Radiant Temperature

✓Radiant temperature:

- Is not important if very similar to the air temperature
- Is at comfortable levels if the difference between DBT and MRT* should not be greater than 3∘Kelvin

(* Mean Radiant Temperature)

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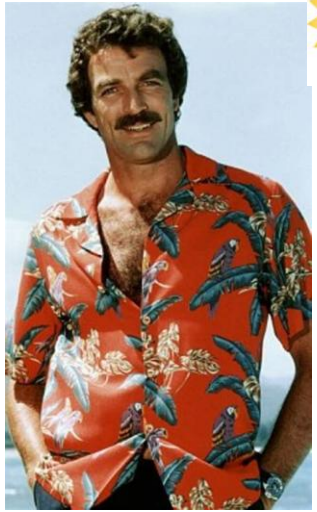

Mean Radiant Temperature

✓The Effect of Mean Radiant temperature (MRT):

- Depends on clothing!!
- In *warmer* climates (people wearing light clothing)
- MRT is about twice as significant as DBT
- Each 1∘C change in MRT is as influential as a 2∘C change in DBT

$$EnvT = \frac{2}{3}MRT + \frac{1}{3}DBT$$

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Achieving Thermal Comfort

Mean Radiant Temperature

✓The Effect of Mean Radiant temperature (MRT):

- Depends on clothing!!
- In *cooler* climates (people wearing heavier clothing)
- MRT has about the same influence as DBT

$$DRT = \frac{1}{2}MRT + \frac{1}{2}DBT$$

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Psychrometric Chart

A chart showing the relationship between air temperature, humidity and moisture content

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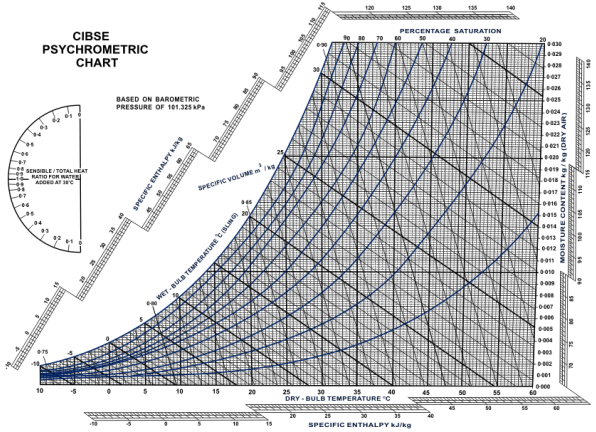
Psychrometric Chart



What is it?

✓The Psychrometric chart:

- A graphic representation to show relationship between air temperature and humidity.
- Includes air temperature (Dry bulb temperature, DBT and wet bulb temperature WBT)
- Includes relative humidity (%RH)
- For a given atmospheric pressure (ATM)

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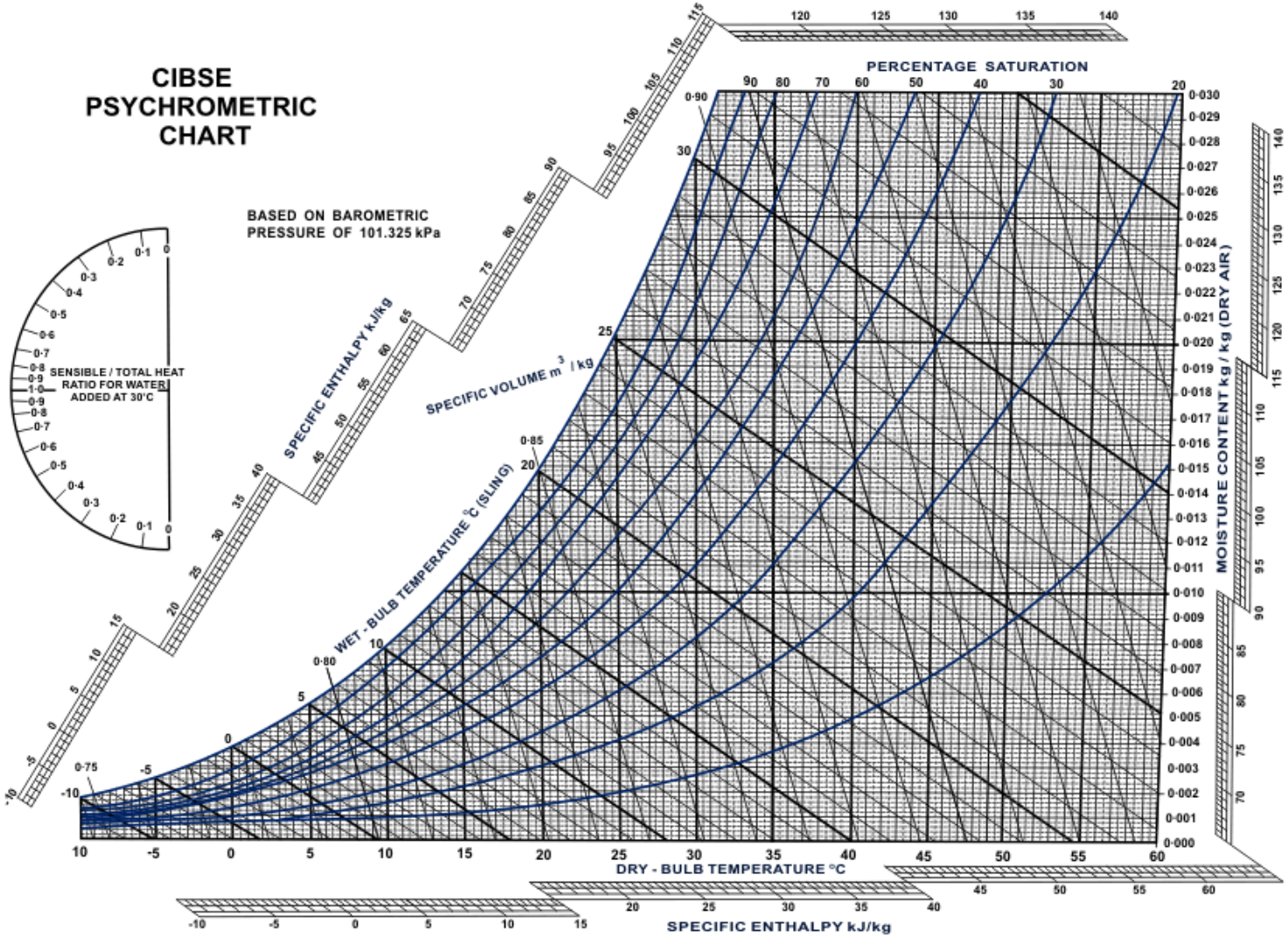
CIBSE PSYCHROMETRIC CHART



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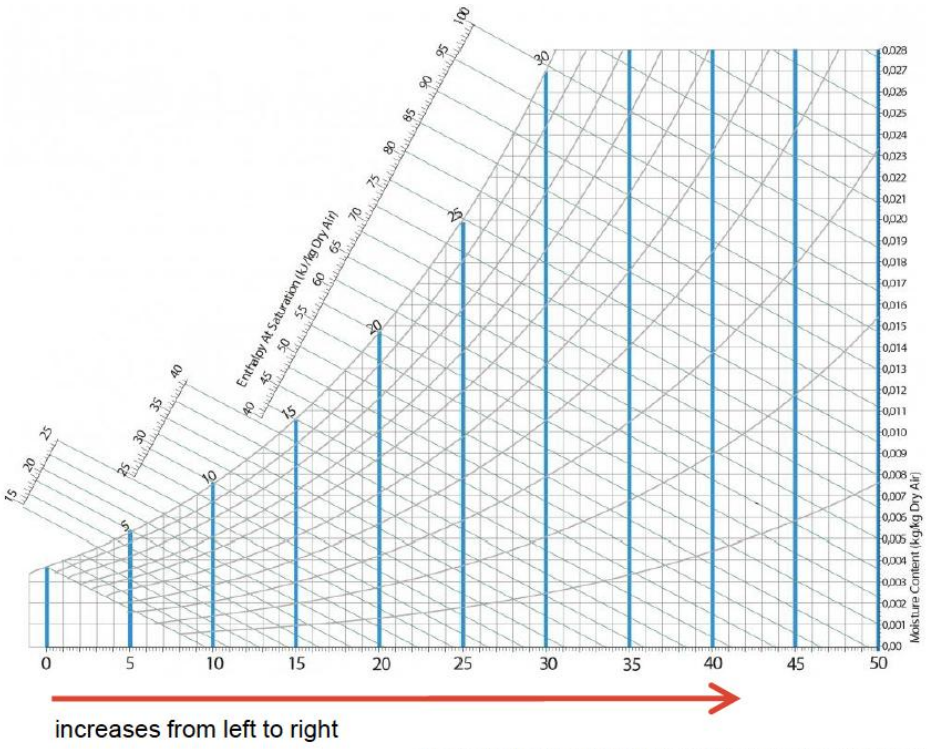
Psychrometric Chart

Dry bulb temperature (DBT)

✓Air temperature



Source: http://academy.autodesk.com/library/building-science/psychrometric-charts

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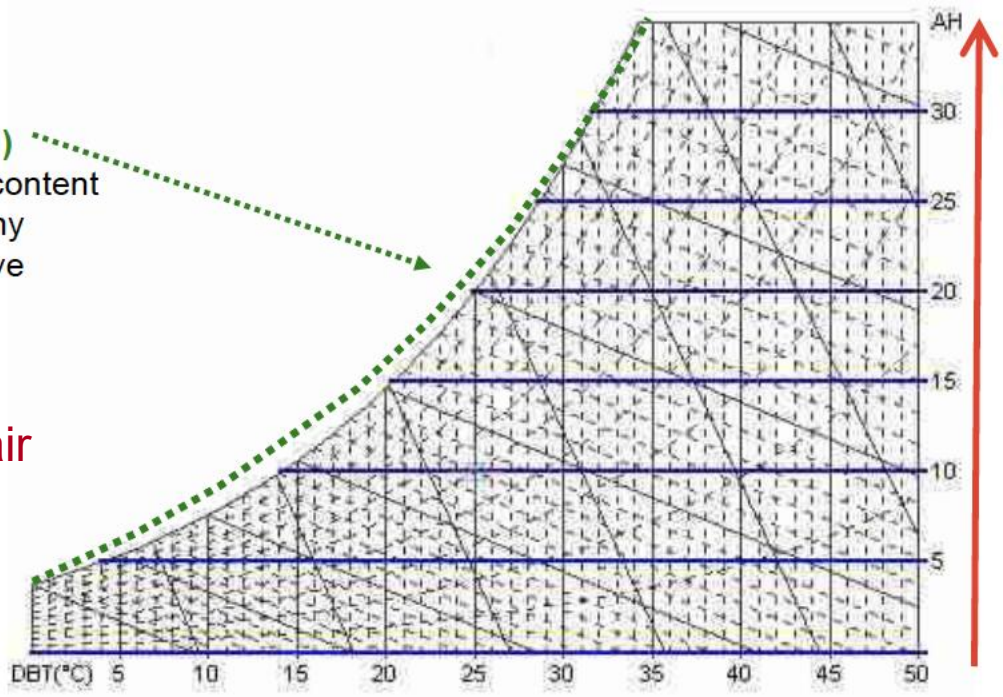
Psychrometric Chart

Absolute humidity (AH)

Saturation humidity (SH)
: the maximum moisture content the air could support at any temperature (100% relative humidity)



✓The water vapour content of air

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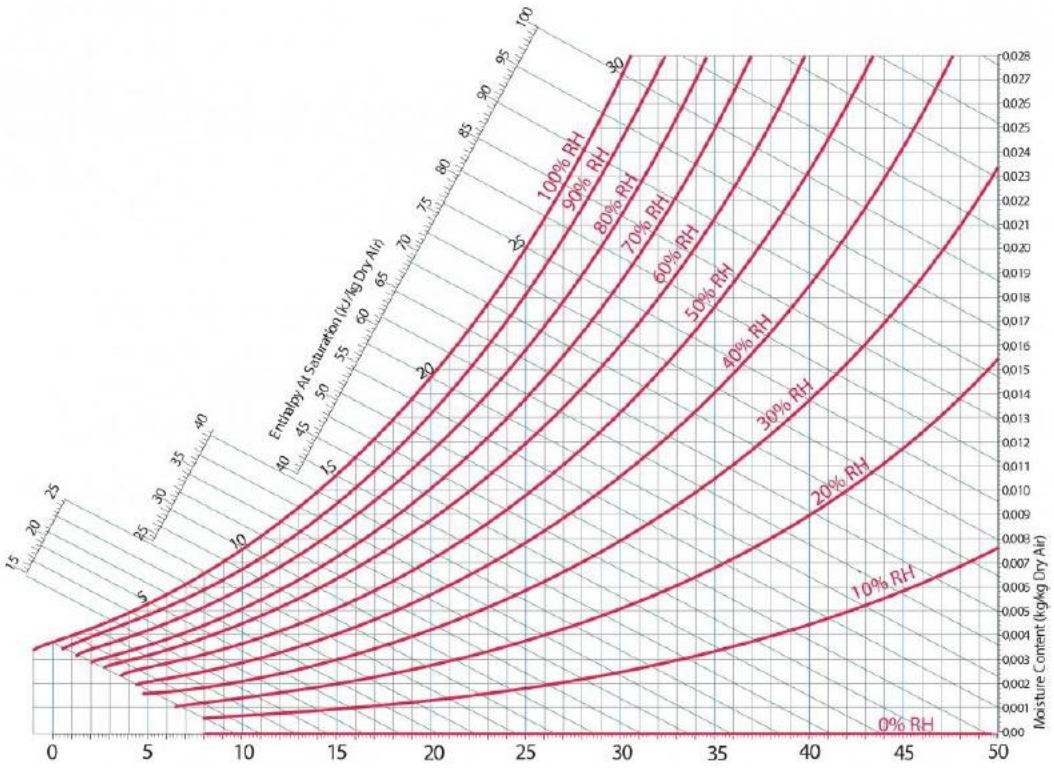
Psychrometric Chart

Relative humidity (%RH)

- ✓RH (%) = AH/SH x 100
- ✓The moisture content of a given atmosphere as a percentage of the saturation humidity at the same temperature
- ✓Begins at 100% to 0% RH



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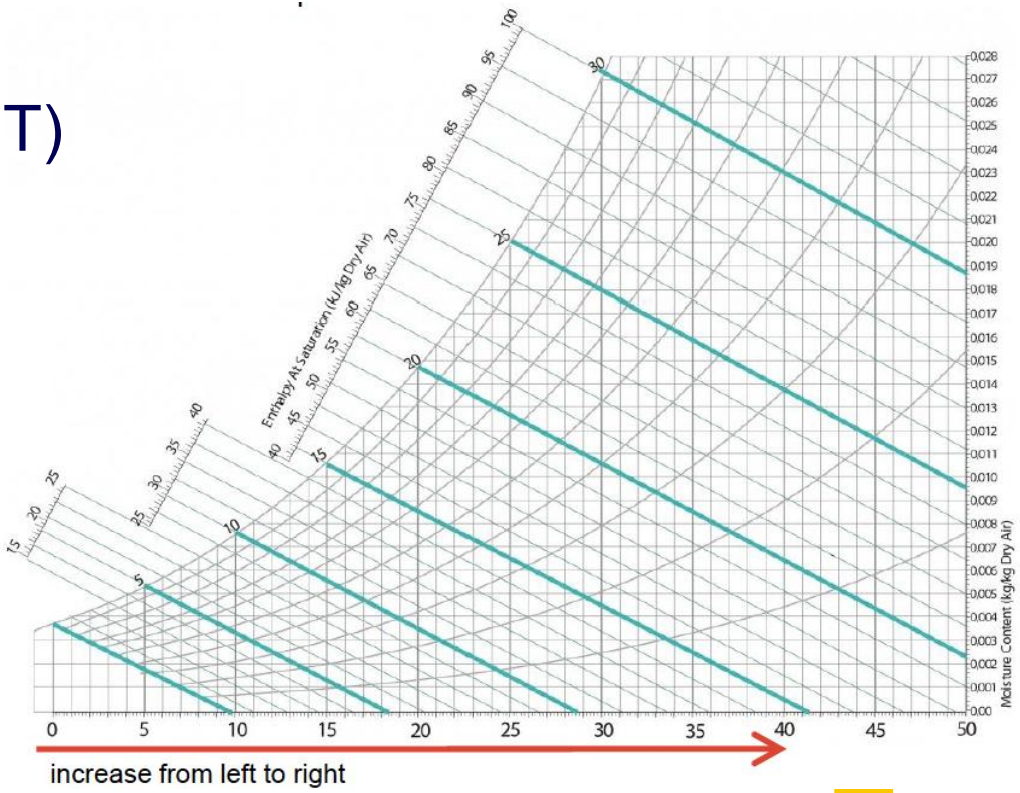
Psychrometric Chart

Wet bulb temperature (WBT)

✓A mixture of air and water vapour



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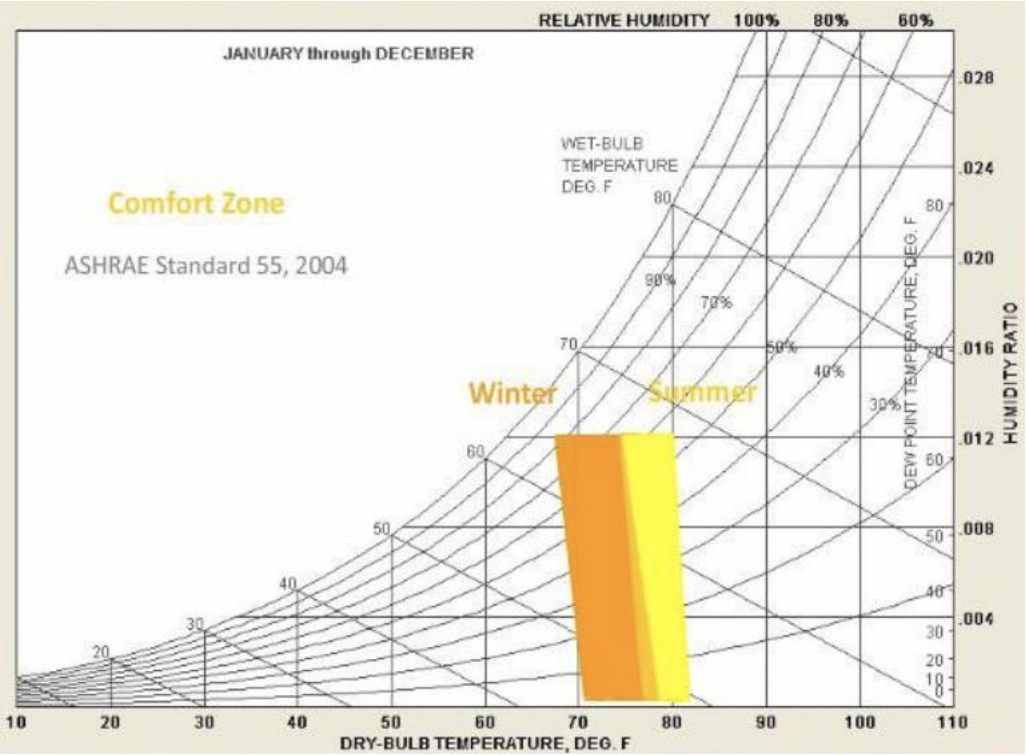
Psychrometric Chart

Comfort zone

- ✓The minimum and maximum temperatures (DB) and moisture content that provide boundaries to a thermal preference



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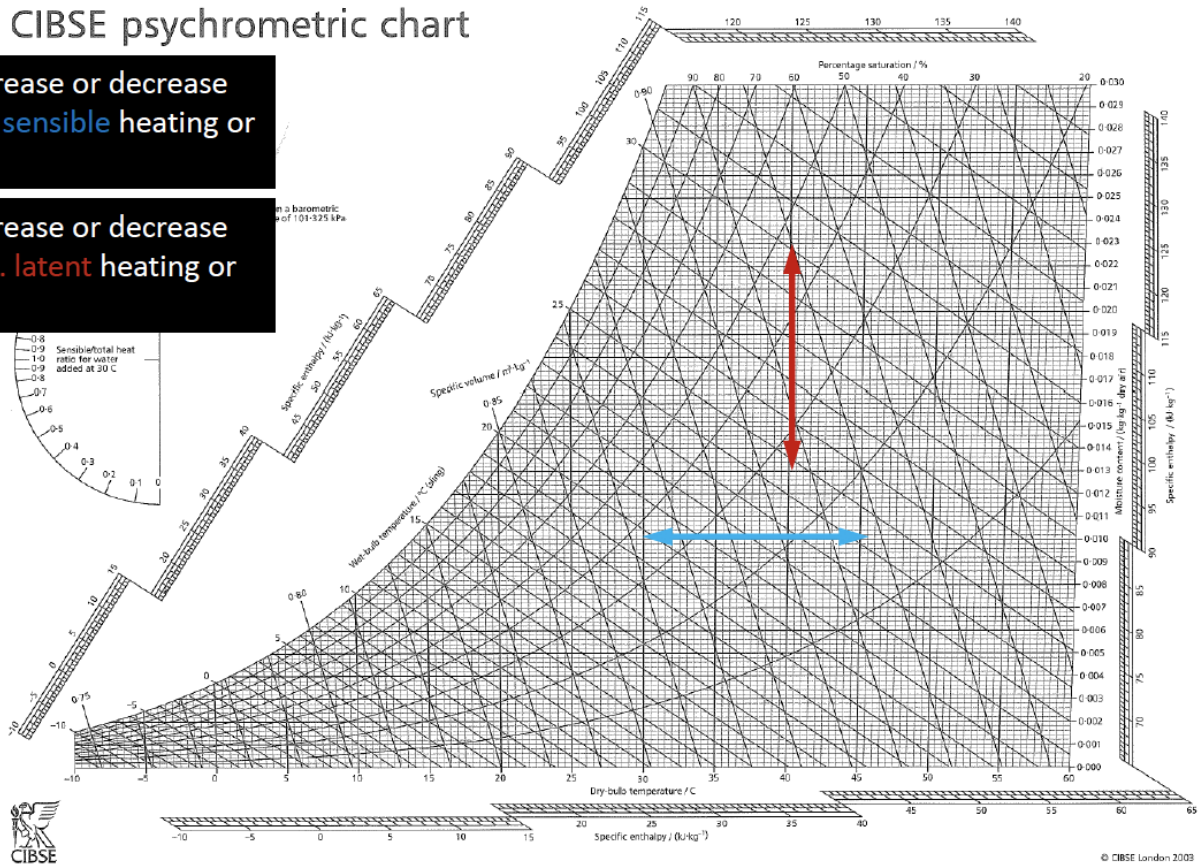
Psychrometric Chart

Sensible & Latent Heat

Equipment may increase or decrease air temperature ie. sensible heating or cooling

Equipment may increase or decrease moisture content ie. latent heating or cooling



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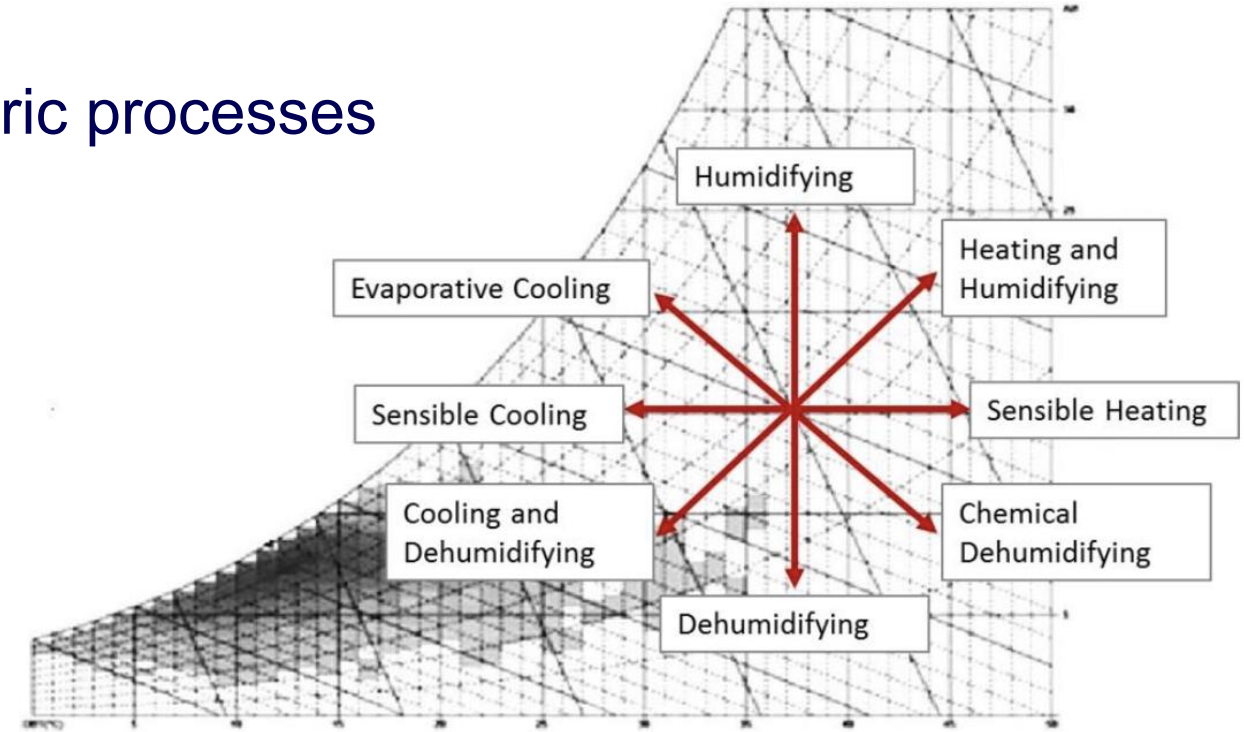
Psychrometric Chart

Psychrometric processes



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Thank you

Next time: Thermal bridges

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